COUNTEREXAMPLES TO NAHM'S CONJECTURE ON THE BLOCH-GROUP TORSION CRITERION FOR MODULARITY

ARGUS OPEN RESEARCH PROJECT

Abstract. We record the counterexamples of Vlasenko and Zwegers (2011) showing that the q-hypergeometric Nahm series $f_{A,B,C}(\tau)$ can be modular for positive-definite rational $r \times r$ matrices A even when the associated element in the Bloch group fails the torsion condition of Nahm's Conjecture 4.1.

1. Main Refutation

Theorem 1.1 (Vlasenko–Zwegers, 2011)**.** *For rank $r \geq 2$, there exist explicit positive-definite symmetric matrices $A \in \mathrm{Mat}_r(\mathbb{Q})$, vectors $B \in \mathbb{Q}^r$, and scalars $C \in \mathbb{Q}$ such that the Nahm series*

$$f_{A,B,C}(\tau) \;=\; \sum_{n \in \mathbb{Z}^r_{\geq 0}} \frac{q^{\frac{1}{2} n^{\mathsf{T}} A n + B^{\mathsf{T}} n + C}}{(q)_n}$$

is a modular function, yet the real solution of the Nahm system $1 - z_i = \prod_{j=1}^{r} z_j^{A_{ij}}$ does not define a torsion element in the Bloch group $\mathcal{B}(\overline{\mathbb{Q}})$.

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